OFFICIAL TECHNICAL RESPONSE & ADDENDUM

Technical Addendum: Infrastructure Hosting Options & Scope Demarcation

Formal engineering response to Al Rabeeh Dental Center's technical inquiry regarding **Vendor-Managed Cloud Hosting**, **Client-Managed In-House Server Specifications (Dell PowerEdge / HPE ProLiant)**, **6-Branch Networking Prerequisites**, and **EMRS Functional Scope Demarcation**.

CLIENT ORGANIZATION	ENGINEERING PARTNER	CLOUD OPTION 1 QUOTE	SCOPE STATUS
Al Rabeeh Dental Center Hamad Town, Muharraq, Isa Town, Budaiya, Galali, Sitra	**SaNDS Lab Middle East W.L.L.** ISO 9001:2015 Healthcare IT Division	**BD 630.000 / Year** Turnkey 3 vCPU / 12GB RAM Managed Cloud	**100% Core Masters Included** Confirmed under BD 8,400.000 baseline

01 Executive Transmittal & Context of this Addendum

Formal Response to Client Inquiry

Purpose of this Document

Following the submission of our Master Commercial & Technical Proposal (Ref: `SL-ARDC-EMRS-2026-V1`), Al Rabeeh Dental Center requested detailed technical specifications, pricing, and architectural requirements for two distinct hosting models: **Vendor-Managed Cloud Hosting** versus **Client-Managed In-House On-Premise Infrastructure** across your 6 Bahrain branches. Additionally, your management requested formal confirmation regarding standard clinic software master modules, workflow governance, and scope boundaries. This addendum provides an exhaustive, fully transparent breakdown to assist your executive steering committee in evaluating both deployment paths.

02 Option 1: Vendor-Managed Cloud Infrastructure (Turnkey Managed Cloud)

Recommended • Zero CapEx • Rapid Multi-Branch Deployment

SaNDS Lab Fully Managed Healthcare Cloud Node

Dedicated high-performance virtual cloud environment pre-configured, hardened, and maintained 24/7 by SaNDS Lab DevOps.

ANNUAL CLOUD INVESTMENT
BD 630.000
Billed Annually • All Inclusive

COMPUTE ENGINE **3 vCPU Cores** High-Frequency Intel/AMD Enterprise Cores	DEDICATED RAM **12 GB RAM** DDR4/DDR5 ECC Cloud Memory	FAST NVME STORAGE **200 GB NVMe** Ultra-Low Latency Database Tier	MONTHLY BANDWIDTH **4 TB Bandwidth** 1 Gbps Unmetered Port Speed
PUBLIC IP ADDRESS **1 Dedicated Static IP** Clean Reputation • SSL Ready	ACCESS LEVEL **Full Root Access** Administered by SaNDS Lab L3 Engineers	CONTROL PANEL **cPanel Included** Commercial License Included	OPERATING SYSTEM **CloudLinux OS** LVE Isolation & CageFS Security
SECURITY SUITE **Imunify360 Included** AI WAF, Malware Scanner & Proactive Shield	OFF-SITE BACKUP **JetBackup + 500GB** Daily Automated Encrypted Off-site Sync	SERVICE HEALTH **24/7 Monitoring** Uptime, MySQL, RabbitMQ & SSL Watch	SYSADMIN SUPPORT **Fully Managed** Zero Internal IT Burden for Al Rabeeh

Why Vendor-Managed Cloud is the Strategic Recommendation

- **Instant Multi-Branch Connectivity:** All 6 branches (Hamad Town, Muharraq, Isa Town, Budaiya, Galali, Sitra) connect securely via standard browser HTTPS (`https://emrs.alrabeeh.com`) without requiring static IP leased lines or costly hardware VPN routers at every clinic.
- **Automated Disaster Recovery:** JetBackup creates daily automated snapshots stored in an independent secondary data center (500 GB storage pool). In case of disaster, point-in-time recovery takes less than 30 minutes.
- **Commercial Simplicity:** The annual fee of **BD 630.000** covers the server, cPanel license, CloudLinux license, Imunify360 enterprise security license, JetBackup license, storage, bandwidth, and SaNDS Lab 24/7 Level-3 system administration.

03 Option 2: Client-Managed On-Premise Infrastructure (In-House Server)

Physical Hardware • Multi-Branch Private Network Prerequisites

Enterprise Physical Rack Server (Dell PowerEdge / HPE ProLiant)

Industrial-grade on-premise hardware deployed at Al Rabeeh Head Office / Central Data Room to host the EMRS backend, database, and queuing engines.

ESTIMATED HARDWARE CAPEX
~BD 3,700 – BD 4,300
One-Time Procurement • Bahrain Market

3.1 Production Server Hardware & Software Dependency Architecture

Component	Recommended Enterprise Specification	Technical Rationale & Sizing Justification
Server Hardware Platform	**Dell PowerEdge R450 / R650** or **HPE ProLiant DL360 Gen11** (1U/2U Rackmount)	Enterprise chassis with hot-swap drive bays, redundant fans, and hardware diagnostic backplane.
Central Processor (CPU)	**Intel Xeon Silver 4410Y / 4416+ (16 Cores / 32 Threads, 2.0–3.9 GHz)** or equivalent **AMD EPYC 9124**	Provides high concurrent thread capacity for 6 clinic branches, PHP-FPM worker pools, RabbitMQ message brokers, and background CRON rollover jobs.
System Memory (RAM)	**64 GB ECC Registered DDR5-4800 MHz (2 × 32GB Dual-Rank)**	Error-Correcting Code (ECC) prevents silent data corruption. 40 GB allocated to MySQL InnoDB Buffer Pool, 16 GB for PHP/RabbitMQ/OS, 8 GB buffer.
Operating System (OS)	**Ubuntu Server 24.04 LTS (64-Bit)** (or Red Hat Enterprise Linux 9 / Rocky Linux 9)	Long-Term Support (LTS) kernel with 5-year guaranteed security patches, optimized systemd daemon management, and native cgroups support.
Web Engine & Middleware	**Apache 2.4 (event MPM) + PHP 8.2 / 8.3 PHP-FPM** + OPcache enabled	High-throughput non-blocking request handling with dedicated fastcgi socket IPC for sub-millisecond response latency.
Database Management System	**MySQL 8.0 Enterprise / MariaDB 10.11 LTS** (InnoDB Engine)	Configured with strict ACID transaction isolation, foreign key constraints, binary logging for point-in-time recovery, and UTF8MB4 collation.
Message Broker Engine	**RabbitMQ Server 3.12+ (Erlang OTP 26)**	Decouples appointment status changes, fail-visit rollovers, and bulk WhatsApp queue processing from frontend HTTP request cycles.
Primary Storage (OS & App)	**2 × 960 GB Enterprise Mixed-Use SATA/NVMe SSD (RAID 1 Mirror)**	Hardware RAID controller (e.g. Dell PERC H755 / HPE MegaRAID with 2GB NV cache) ensures zero downtime in case of a drive failure.
Database Storage (MySQL)	**2 × 1.92 TB Enterprise Read-Intensive/Mixed-Use NVMe SSD (RAID 1)**	Dedicated hardware RAID 1 volume separated from OS disk. High sustained random write IOPS (>80,000 IOPS) prevents database lock contention.
Network Interface (NIC)	**Dual-Port 1 Gbps / 10 Gbps RJ45 Ethernet (LACP Teaming / Failover)**	Bonded network interfaces provide link redundancy and aggregated bandwidth for concurrent multi-branch queries.
Power Supply (PSU)	**Dual Redundant Hot-Plug Power Supplies (1+1 Platinum, 800W+)**	Connected to separate electrical circuits / UPS lines for true electrical fault tolerance.

Component	Recommended Enterprise Specification	Technical Rationale & Sizing Justification
Out-of-Band Remote Mgmt	**Dell iDRAC9 Enterprise** or **HPE iLO 6 Advanced**	Dedicated isolated network port allowing remote BIOS setup, virtual KVM console, OS re-installation, and remote power cycling.
Dedicated Local Backup	**Synology DiskStation DS923+ / RS822+ 4-Bay NAS (4 × 4TB IronWolf Pro in RAID 5 = 12TB usable)**	Automated nightly snapshot repository separated physically from the main server; stores daily MySQL dumps and patient attachment archives.

3.2 Estimated Hardware Procurement Costs in Bahrain / GCC (One-Time CapEx)

Procurement Item Reference	Estimated Cost (USD)	Estimated Cost (BHD)	Commercial Status
Dell PowerEdge R450/R650 Server: Intel Xeon Silver 16-Core, 64GB ECC, 2×960GB SSD, 2×1.92TB SSD, PERC H755 RAID, Dual 800W PSU, iDRAC Enterprise, 3-Yr ProSupport NBD	$6,500 – $7,300	**BD 2,450.000 – BD 2,750.000**	Purchased by Client directly from Dell Authorized Partner in Bahrain
Dedicated Network Backup NAS: Synology RS822+ / DS923+ 4-Bay NAS with 4 × 4TB Enterprise NAS SATA HDDs (RAID 5)	$1,450 – $1,720	**BD 550.000 – BD 650.000**	Purchased by Client from Bahrain IT Distributor
Enterprise Rackmount Smart-UPS: APC Smart-UPS 2200VA / 1980W LCD 230V with Network Management Card (AP9640)	$1,200 – $1,460	**BD 450.000 – BD 550.000**	Essential to protect database SSDs from sudden power outage write-corruption
Rack Cabinet & Infrastructure: 18U–24U enclosed sound-dampened server cabinet with PDU, patch panels, cable trays, and rack shelving	$660 – $930	**BD 250.000 – BD 350.000**	Physical security and structured cabling at HQ data closet
TOTAL ESTIMATED INITIAL HARDWARE CAPEX (PROCURED BY CLIENT)	**$9,810 – $11,410**	**BD 3,700.000 – BD 4,300.000**	**One-Time Capital Investment (3 to 5 Year Asset Life)**

3.3 Critical Infrastructure Prerequisites for Connecting Your 6 Branches

If the central server is hosted in-house at your Head Office or Main Clinic, the remaining 5 branches must be able to securely query and update patient files in real time. The following physical and networking components are **mandatory** to ensure continuous uptime and NHRA data protection:

Central Server (HQ) | Hamad Town Branch | Muharraq Branch | Isa Town Branch | Budaiya Branch | Galali Branch | Sitra Branch

1. HQ Dedicated Business Fiber + Static IP

A high-reliability Business Fiber connection (minimum 50–100 Mbps symmetrical) with a permanent **Public Static IPv4 Address** from Batelco, Zain, or stc Bahrain is required at the central server location (~BD 80 – BD 150 / month). Residential or dynamic broadband lines cannot be used.

2. Central Next-Gen Enterprise Firewall

A hardware Next-Generation Firewall (e.g. **Fortinet FortiGate 60F/70F** or **Sophos XGS 116**) deployed at HQ (~BD 450 – BD 650) to manage intrusion prevention (IPS), DoS attack mitigation, SSL inspection, and IPsec VPN tunnel termination from the branches.

3. Secure Site-to-Site IPsec VPN Tunnels

Recommended Architecture: Deploying managed VPN routers (e.g. FortiGate 40F or MikroTik hEX) at each of the 6 branches creating encrypted Site-to-Site IPsec VPN tunnels. Reception workstations access the server over a private internal subnet (e.g. `https://192.168.10.50`), completely hidden from the public internet.

4. 24/7 Climate Control & Physical Security

The physical server must be housed in an enclosed, lockable server rack cabinet within an air-conditioned room running continuously at 18°C–22°C (24/7/365). Power cuts or A/C failures in Bahrain summers can cause catastrophic thermal throttling or hardware shutdowns.

5. Smart CPR Hardware LAN/WAN Integration

The custom SaNDS Lab .NET CPR Card Windows Service installed on each reception workstation communicates with the central server API over the VPN/LAN. Stable network latency (<25ms) is required between all branches and HQ for instantaneous patient retrieval.

6. Ongoing SysAdmin & Disaster Recovery

Client must designate an internal IT administrator (or retain a managed IT support vendor) responsible for monitoring daily NAS backups, applying Ubuntu OS security patches, testing UPS battery degradation, and swapping failed SSD drives.

04 Strategic Comparison Matrix: Cloud vs. In-House Hosting

Architectural Decision Guide

Evaluation Dimension	Option 1: Vendor-Managed Cloud (SaNDS Lab)	Option 2: In-House On-Premise Server (Al Rabeeh)
Upfront Capital Expenditure (CapEx)	**BD 0.000 (Zero Upfront CapEx)**	**BD 3,700 – BD 4,300** (Server, NAS, UPS, Rack)
Annual Recurring Operating Cost (OpEx)	**BD 630.000 / Year** (All-Inclusive)	**~BD 1,200 – BD 1,800 / Year** (Static Fiber, UTM License, Hardware AMC)
3-Year Total Cost of Ownership (TCO)	**BD 1,890.000 Total** (3 Years × BD 630)	**~BD 7,300 – BD 9,500 Total** (CapEx + 3 Years OpEx)
Deployment Time to Go-Live	**24 to 48 Hours** (Instant Cloud Provisioning)	**4 to 6 Weeks** (Hardware procurement, import, rack installation & cabling)
Connecting 6 Remote Branches	**Seamless via Secure HTTPS** (Any internet connection works)	**Requires Site-to-Site VPNs**, static IP at HQ, and hardware firewall
Disaster Recovery & Off-site Backup	**Automated Daily Off-site** (500GB JetBackup repository)	**Local NAS backup**; off-site cloud sync requires extra storage subscription
Security, WAF & Anti-DDoS	**Enterprise Imunify360 + AI WAF** + CageFS Included	**Requires FortiGate / Sophos** UTM annual security licensing
Power Cut & Internet Failure Risk	**Tier-3 Data Center (99.9% Uptime)** with dual generators	**High Risk if HQ Power/Fiber Fails** (All 6 branches lose access simultaneously)
Maintenance & SysAdmin Burden	**100% Handled by SaNDS Lab L3 DevOps**	**Requires Dedicated In-House IT** or separate IT support contract
Strategic Recommendation	**STRONGLY RECOMMENDED FOR AL RABEEH**	**FEASIBLE IF CLIENT HAS EXISTING DATA CENTER**

05 Scope Demarcation & Responses to Client Clarifications

Complete Alignment with Client Expectations

We appreciate Al Rabeeh Dental Center's clear articulation regarding standard clinic management functionality. Below is our definitive, legally binding scope demarcation:

100% INCLUDED in the Existing Proposal (BD 8,400.000)

Covered Scope

- **Master Creation & Management:** Complete administrative masters for Clinics, Branches (all 6 branches), Treatment Rooms, Dental Chairs, Services, Procedures, ICD-10 Diagnostic Codes, ADA/FDI Dental Codes, and Price Lists.
- **User & Doctor Management:** Comprehensive profiles for Receptionists, Nurses, General Dentists, Orthodontists, Specialists, Branch Managers, Cashiers, and Admins.
- **Role-Based Access Control (RBAC):** Granular permissions matrix ensuring doctors view clinical workflows, receptionists manage queues and bookings, and admins control approvals.
- **Standard Validations & Clinical Logic:** Overlapping appointment locks, doctor duty schedule enforcement, Bahrain CPR 9-digit format check, 8-digit mobile phone validation, and mandatory field guards.
- **Confidentiality, Security & Audit Trail:** AES-256 database encryption at rest, HTTPS SSL in transit, immutable audit logs tracking user, timestamp, IP, and before/after values for all records.
- **All 11 Blueprint Modules:** Dashboard & 6 KPIs, Multi-Doctor Calendar, MSD Leakage Tracker, Fail Visit Rollover, Ortho Continuity, TP Value Recall, Feedback Calls, Lab Tracker with Shade Master, Insurance Claims, Patient Journey Compliance, Billing Cashier Register.
- **Continuous Stage-by-Stage Guidance:** SaNDS Lab technical delivery leads will actively walk through each milestone demo and UAT session, guiding Al Rabeeh on configuration, master data population, and workflow optimization.

Items Requiring Separate Discussion & Quotation

Out of Scope

- **Pharmacy ERP & Full Inventory Purchasing:** Full enterprise pharmaceutical stock replenishment, supplier purchase orders, and expiry batch dispensing (basic clinic procedure consumable linking is included, but full pharmacy purchasing ERP is separate).
- **Automated Laboratory Machine Hardware Interfacing:** Bidirectional HL7/ASTM robotic interfaces to external automated blood or diagnostic analyzers (the EMRS includes manual lab slip entry and 3-lane chat).
- **In-House Radiology / PACS DICOM Archive:** The EMRS links to digital X-ray image files and external URLs; a dedicated hospital-grade PACS DICOM archive server is not included.
- **Third-Party Messaging Gateway Fees:** Official WhatsApp Business API per-template utility charges or bulk SMS SMS-gateway credits (billed directly by Meta / Batelco / Zain at actual consumption).
- **Third-Party TPA API Licensing Fees:** Any proprietary API connector charges levied directly by insurance portals (Globemed, Mednet, Gulf Union) for automated EDI transmission.
- **Revisions Exceeding the 3-Revision Cap:** Iterations on the same feature exceeding three rounds will be handled via standard 10% Milestone Change Requests (CR) as agreed in governance rules.

Clarification on Post-Deployment Support & Warranty

Regarding the client's reference to *"one year of support"*:

- **Included in BD 8,400 Contract:** Milestone `MS-BUF` provides **1 Full Month (4 Working Weeks) of Multi-Branch Pilot & Hypercare Support**, followed by our standard **60-Day Post-Go-Live Comprehensive Warranty** covering bug fixes, defect rectification, and performance tuning at zero additional cost.
- **Extended 1-Year Comprehensive SLA:** Following the warranty period, ongoing annual Level-2 and Level-3 application support (Annual Maintenance Contract / AMC) can be seamlessly bundled with the **Vendor-Managed Cloud Hosting** or engaged as an annual SLA agreement at standard industry rates (15% of software value / year).

06 Formal Acknowledgment & Next Steps

✓ Ready for Executive Approval

Please confirm your selected hosting model (**Option 1: Vendor-Managed Cloud at BD 630/Year** or **Option 2: Client In-House Infrastructure**) by signing below or returning confirmation via email. Upon selection, SaNDS Lab will initiate the environment provisioning and schedule the project kickoff sprint.

For Al Rabeeh Dental Center (Client)

Authorized Signatory:

_____

Designation / Title:

_____

Selected Hosting Option: [ ] Option 1 (Cloud) [ ] Option 2 (In-House)

Signature & Official Seal Date: ____ / ____ / 2026

For SaNDS Lab Middle East W.L.L. (Partner)

Authorized Signatory: Technical Delivery & Solutions Directorate

Engineering Division: Healthcare Information Systems

Proposal Ref: SL-ARDC-EMRS-INFRA-2026-V1

Authorized Signature Date: 04 November 2026